

## Installation Instructions ADSS® All-Dielectric Self-Supporting Distribution Line

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## ADSS Installation Instructions

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### 1.0 GENERAL INFORMATION

The following instructions provide general information for engineers, managers, and construction personnel on the methods for placement of all-dielectric, self-supporting (ADSS), fibre optic cable specific to electric distribution system environments. Please contact AFL for installation of ADSS fibre optic cable in higher voltage transmission lines (>69kV) and for further information such as minimum separation and clearance, sag tables, and specific cable data which is not contained in this document.

Installation equipment or methods not mentioned in this practice are not approved for use with ADSS, and its use without specific approval by AFL shall be at the risk of the customer. Please contact AFL with questions about any particular method or equipment recommendations.

### 2.0 SAFETY

AFL strongly recommends the use of approved personal protection equipment as needed per installation.

All protection equipment must meet the relevant safety requirements and all personnel involved should be aware of all applicable Safe Work Australia regulations, state and local regulations, and company policies.

### 3.0 INCOMING CABLE INSPECTION

Before starting construction, use an optical time-domain reflectometer (OTDR) to verify that no damage to the fibre has occurred during shipment. The cable will have a final fibre test report on each fibre from manufacturing that is included on the reel before shipment. Contact AFL for electronic copies of the final fibre test report. Visually inspect the reel and cable for damage as well.

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### 4.0 CABLE PRECAUTIONS

Review the ADSS Fibre Optic Cable specification sheet before installation, which includes a 'Quick Reference Installation Notes' section. Violating cable precautions can affect the optical transmission capability of the fibre, in turn affecting network integrity and may require cable replacement.

#### 4.1 Minimum Bend Radius

Ensure that the cable is not kinked and that the minimum bend radius is not exceeded. ADSS cable has a minimum bending radius value for dynamic (under load) and static (no load). Do not sag/tension the cable around large angles as this will cause a high crushing force to be placed on the cable at the angle sheave, which can damage the fibres. Ensure that the minimum stringing sheave diameter is met, refer to 'Sheaves' sections below.

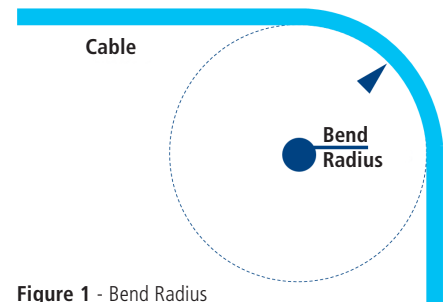


Figure 1 - Bend Radius

#### 4.2 Maximum Tension

There are two tension values to monitor during installation: stringing tension (at tensioner) and the span initial sagging tension. Tension the cable from the dead-end segment to dead-end to avoid tensioning around angles greater than 30 degrees. Sagging individual spans at tangent locations can create an unbalanced load and cause the cable to slip through the tangent hardware. The maximum recommended stringing tension is 20% of the cable breaking strength. Control the rotation of the payoff reel to prevent overrunning and apply only a minimal amount of braking. Braking should be applied to the reel through the support shaft, and not by methods like wedging a wood block under the reel flange.

The span initial sagging tension is typically based on 1% to 1.5% vertical sag of the span distance. Wire mesh grips are intended only for pulling the cable through the system, do not use wire mesh grips to tension or to hold the cable under tension. Use only approved gripping and pulling devices when tensioning, or temporarily holding fully tensioned ADSS cable. See the 'Tensioning Grips' section below.

#### 4.3 Crush and Twist

Do not allow vehicles to run over the cable. At road crossings, the cable should be suspended above roads, driveways, etc. during installation. The placement of clips on a temporary slack span of rope or previously installed cables can suspend the cable above road crossings. Do not allow the cable to twist as it is pulled through sheaves. If a continuous twist in a constant direction is observed, stop the installation, ease off the tension, and readjust the sheave. Due to the light weight of ADSS and the relative low stringing tensions, the sheave may require support at the base to prevent the cable from riding out of the sheave or twisting during installation (Figure 2).



Figure 2 - Sheave A

#### 4.4 Abrasions

Before starting construction, survey the proposed cable route to assure that the right-of-way is clear of obstructions that may interfere with the installation. Do not drag the cable over obstacles in the span or on the ground. It is recommended that if obstructions occur, they should be removed, or use a series of hold-down blocks to prevent contact with the obstruction.

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### 4.5 Cutting Cable

Do not cut the cable without prior approval of the construction supervisor or project engineer or other authorised associate. Cable cuts will add splices, which will add cost and optical degradation that is not accounted for in the overall optical loss budget. The number and location of splices are usually determined in the initial system design.

### 4.6 Temporary Ground Slack

A 'figure-8' configuration should be used when ADSS cable is placed on the ground in lengths of greater than 30 m. The figure-8 size should not violate the dynamic bending radius.

## 5.0 INSTALLATION METHODS

The two primary methods for the placement of ADSS fibre optic cable are; the stationary reel, sometimes called the 'Back-pull Method', and the moving reel, sometimes called the 'Drive-off Method.' The drive-off method is acceptable but is generally used only when the cable is the lowest on the structure or in a new build situation. The stationary reel method is more common.

### 5.1 Stationary Reel: 'Back-pull Method'

The stationary reel method of cable installation most useful for application on distribution lines, where there are many obstacles, such as lateral branches or taps (Figure 3).

The cable reel is placed on a reel stand or reel trailer, supported by the arbor holes, at a fixed location. A braking device applies minimal tension to the reel to prevent overrun. The reel stand should be placed in-line with the first two structures to avoid twisting of the cable or any abrasion to the cable by rubbing on the sides of the sheave groove. A pulling slope of four horizontal to one vertical is considered good practice.

Holes are drilled, and machine bolts (or similar hardware) are mounted to the structure at the appropriate mounting height. At dead-end and tensioning locations, down-guys of an appropriate loading factor are placed if necessary.

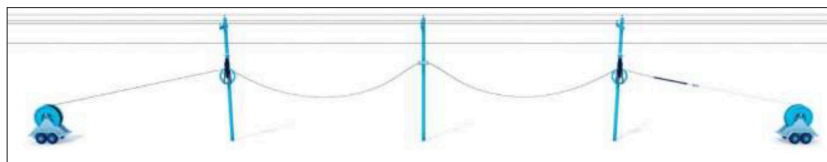


Figure 3 - Stationary Reel

Sheaves or tangent trunnion hardware are placed just above or below the desired cable framing location of each support structure.

Small pilot lines run through the sheaves at each support structure. The pulling line is pulled from the pulling location back through each sheave using the pilot lines. After the pulling line is fed through the entire section to be pulled, it is attached to the ADSS cable with a break-away swivel link rated for the cable and a wire mesh grip. See the 'Tensioning Grip' section below for details.

The ADSS cable is then pulled through the entire section with the puller and tensioner. Several pulling stages may be required to place the cable through the whole system.

With the cable deployed, starting at an end location, each dead-end to dead-end cable segment can be sagged and tensioned, and support hardware applied according to the installation requirements. Sagging and tensioning must be from dead-end to dead-end, as sagging individual spans at tangent locations can create an unbalanced load and cause the cable to slip through the tangent hardware.

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### 5.2 Moving Reel: 'Drive-off Method'

The moving reel method is not utilized as often as the stationary reel method. Its primary application is in the construction of new lines with clear right-of-way and no obstacles (Figure 4).

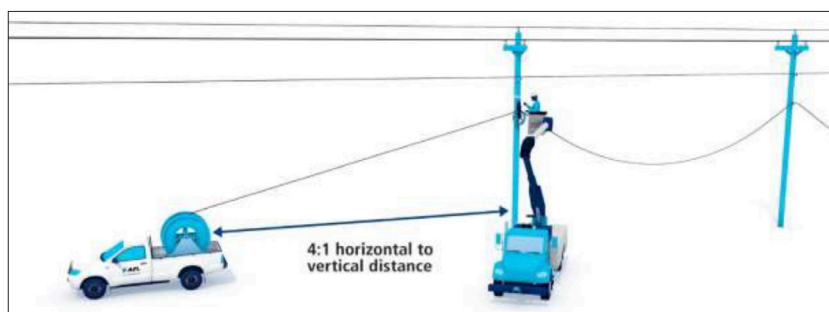


Figure 4 - Moving Reel

Place the cable reel in a reel trailer or line truck equipped with reel carrier, supported by the arbor holes. The cable should payoff the top of the reel from the back for reel trailers. A braking device is used to prevent overrun of the reel when stopping at the support structures.

The reel stand should be placed in-line with the first two structures to prevent twisting of the cable or any abrasion to the cable by rubbing on the sides of the sheave groove. A pulling slope of four horizontal to one vertical is considered good practice.

Holes are drilled, and machine bolts (or similar) are placed on the structures at the appropriate mounting height. At dead-end and tensioning locations, down-guys of an appropriate loading factor are placed if necessary.

The sheave or tangent trunnion hardware is placed above or below the desired framing location of each support structure, and the cable is dead-ended at the starting location.

With minimal tension applied to the reel brake, the reel of cable is transported along the construction route, and the cable is paid out. As the reel empties, the back tension will have to be periodically adjusted to account for the difference in reel weight.

As the moving reel passes a support structure, the pulling is stopped, and the cable is placed into the sheave attached to the structure at the desired framing height.

The reel proceeds to the next support structure, where the process is continued until the cable is completely deployed.

With the cable deployed, starting at an end location, each dead-end segment can be sagged and tensioned, and support hardware applied according to the installation requirements. An alternative procedure is to sag and tension each span and install permanent hardware as the cable is being deployed. In either method, sagging and tensioning must be from dead-end to dead-end, as sagging individual spans at tangent locations can create an unbalanced load and cause the cable to slip through the tangent hardware.

### 5.3 Sag and Tension

Sagging and tensioning of a system is worked progressively from one end of the system towards the opposite end. Typically, the cable slack is worked back toward the payoff reel to recover as much cable as possible. This should always be done in dead-end segments and never span to span at non-dead-end locations.

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### 5.3.1 Termination Point

Following local practices, pull enough additional cable to reach the termination location and to facilitate cable splicing. The spare fibre required in the splice tray is dependent on the type of fibre organizer and splicing method. Typically, one to two metres of fibre is required to splice in addition to slack storage at the location physically. Assume that 4.5 m of cable is cut off at the wire mesh grip end as fibres in this section can be damaged during installation.

### 5.3.2 Termination Structure

At the last structure, establish a dead-end assembly. Assume that the bend radius requirements are maintained where the cable goes down the structure. Riser guards are recommended for ADSS cable going down the structure.

Remove all excess cable slack out of the series of spans. This step is not pre-stressing or even tensioning and is necessary to position the temporary dead-end pulling grip properly. To remove the slack, reverse the tensioner and pull the cable back toward the reel, being careful not to exceed the pulling criteria of 50% of the maximum installation tension. Do not attempt to tension the cable using the tensioner and the cable on the reel as this may crush the cable.

With the cable slack removed, apply a temporary dead-end or AFL temporary grip assembly 1.5 to 2 dead-end assembly lengths (approximately two to three metres) from the structure. This device will serve as a tensioning grip for proper span sag and tension before installing the permanent dead-end. Attach the tensioning rig, comprised of a sufficiently rated chain

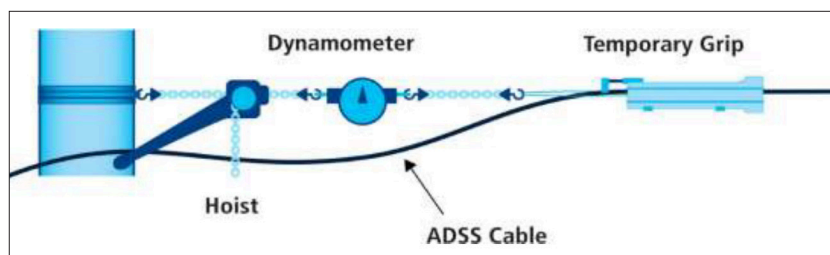


Figure 5 - Tensioning

Typically, the cable is worked dead-end to dead-end segment back to the payoff reel, while trying to avoid angles greater than 30 degrees (which require a dead-end). After the spans are at proper sag and the dead-ends attached, the suspension or tangent hardware is installed and attached to the structures by working back to the dead-end one span at a time. Care should be taken to ensure when installing the suspension or tangent hardware the sag is not altered based on what was set by tensioning from dead-end to dead-end.

Once the permanent dead-end is installed and the hardware is attached to the structure, the tension can be released on the chain hoist and the temporary dead-end removed. As the next permanent dead-end is installed on the adjacent span, make sure that the expansion loop (if required) under the dead-ends is appropriately formed, maintaining the cable minimum bend radius. This process is repeated until all spans are sagged and tensioned for the complete system.

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### 6.0 INSTALLATION EQUIPMENT

ADSS fibre optic cable is typically supplied on wood reels. The cable reel includes a protective covering, and sometimes wood lagged to provide additional protection during transportation. Steel reels available upon request as needed. Contact AFL for further information on wood and steel reels, reel dimensions, and the overall reel weight. For details on reel handling, refer to the separate document 'Cable Storage and Handling Guide.'

#### 6.1 Reel Stands

Reel stands are designed to be used with tensioners to supply the necessary hold-back tension to the cable. The stand should be selected to accommodate the cable reel dimensions and gross weight. AFL standard reels are not designed to withstand the forces developed by braking during high tension stringing operations. The cable can be pulled directly from the reel stand only when employing slack stringing methods that allow minimal tension to be applied directly to the reel of cable. Direct tension stringing from the reel at cable installation stringing tensions should not be attempted.

#### 6.2 Pulling Machines

Both bull wheel and reel type pulling machines may be used to install ADSS fibre optic cable. Availability and previous experience with a pulling machine should be factors when determining the type of pulling machine to be used.

#### 6.3 Pulling and Tensioner Operating Characteristics

The pulling and braking system should be operated smoothly to prevent any sudden jerking or bouncing of the cable during deployment. Pullers and tensioners should be equipped with tension indicating and limiting devices. The selection of the puller and tensioner shall be dependent on the stringing tension, the actual cable weight, and length to be installed. Positive braking systems are required for pullers and tensioners to maintain cable tension when pulling is stopped. Fail-safe type braking systems are recommended.

#### 6.4 Sheaves

The terms sheave, traveler, pulley, and block are used interchangeably for the same piece of installation hardware. This document primarily uses 'sheave.'

##### 6.4.1 Sheave Diameter

The diameter of the sheave should not be less than 300 mm at mid-span suspension points. Some tangent hardware can be used in place of the sheave (e.g., AFL trunnion). For line angles (horizontal or vertical) greater than 25° and the first and last position in the pull (generally parameters for dead-end structures), the minimum diameter of the sheave should be 40 times the cable diameter.

Example:

Cable Diameter = 11 mm

40 x 11 mm = 440 mm

Sheave Diameter above 25°/First and Last Structures = 440 mm

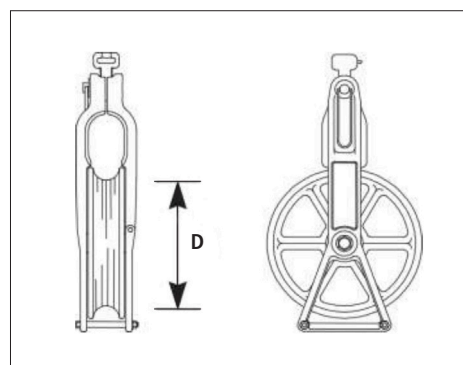


Figure 6 - Sheave Diameter

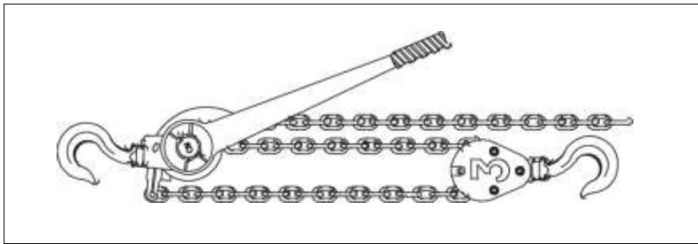
##### 6.4.2 Sheave Material

The sheave should be in good working order and properly lubricated. It is recommended that the sheave has an elastomeric liner that will provide cushioning and minimize any abrasion to the cable jacket, linings of neoprene or urethane are acceptable. The liner should not be torn or loose.

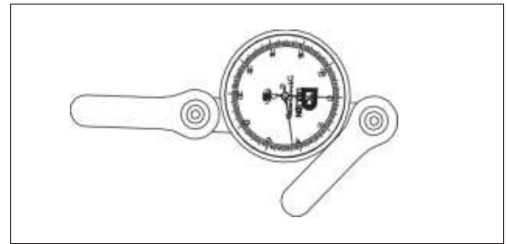
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### 6.5 Chain Hoist

Chain hoists are used to tension and sag each dead-end cable span (**Figure 7**). The hoist should be rated for the maximum installation load of the ADSS cable, plus an appropriate safety factor. The hoist shall be in good working order and adequately maintained. Ensure the chain is not deformed, twisted, or corroded. Inspect release levers and cam action for proper operation.



**Figure 7** - Chain Hoist



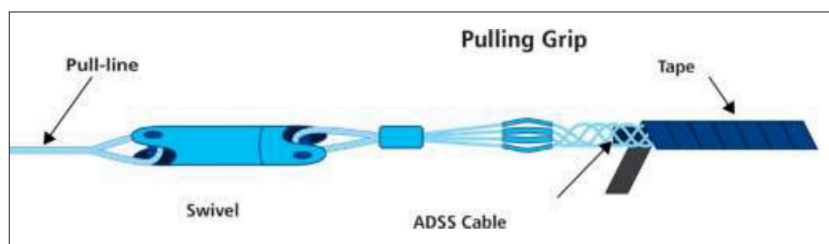
**Figure 8** - Dynamometer

### 6.6 Dynamometers

Dynamometers are used to measure the tension at each dead-end-dead-end segment (**Figure 8**). The dynamometer should be rated above the maximum installation load of the ADSS cable. Dynamometers should be used in conjunction with sight sagging to ensure proper tensioning of the cable.

### 6.7 Pulling Grips

Wire mesh grips are used to pull the fibre optic cable through the sheaves. The grip should be a double or triple weave design and appropriate for the cable diameter. The load rating shall match the maximum anticipated load on the cable during cable pull-in, which is typically well under the sagging tension. The grip should have a break-away swivel link rated to the maximum installation tension that will minimise cable twisting and protect the cable (see **Figure 5** above). Grips should be applied per the manufacturer's instructions. Do not use pulling grips to tension the ADSS cable.



**Figure 9** - Pulling Grip



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### 6.8 Tensioning Grips

A separate tensioning grip than wire mesh pulling grips must be used to temporarily grip the cable during the sagging and tensioning. This grip can either be a formed wire dead-end specified for the cable, or AFL's 'Temporary Tensioning Grip' (Figure 10). If formed wire grips are used, follow the re-use instructions. Typically, formed wire grips may be re-used no more than two times.



Figure 10 - Temporary Grip

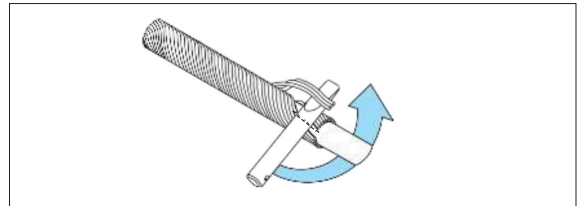


Figure 11 - Formed Wire Installation Tool

### 6.9 Formed Wire Installation (FFT) Tools

The use of a Formed Wire Installation Tool (Figure 11) is recommended to reduce the opportunity of damage to the cable jacket that can be caused by screwdrivers. The use of screwdrivers is not recommended.

## 7.0 INSTALLATION CONSIDERATIONS

### 7.1 Reel Preparation

AFL ships the cable reels with the inner tail securely connected to the outside of the reel flange. Before stringing, this connection should be loosened, but not removed. This process allows the inner layers of cable to adjust themselves to the varying tensions seen during installation. As the cable makes these adjustments, the inner tail may lengthen ('grow'), requiring periodic attention to ensure that the cable continues to be in a state where it can 'grow' out. Tighten reel flange bolts before cable installation as bolts can loosen during shipment and storing. Cable crossover during installation can occur if wood flanges are warped due to long storage outside or the flange bolts that are not tightened. Decrease the pulling tension if the cable does not payoff smoothly.

### 7.2 Splicing

At the locations that fusion splicing of fibres is required, additional slack cable must be provided to accommodate the splicing process. AFL recommends splicing on the ground and not in an aerial bucket. A splice vehicle may be required, and enough cable will be necessary to reach the vehicle. Ensure that the 4.5 m of cable to be removed from the gripped section is accounted for in the slack.

After the excess cable length is determined, coil the cable (maintaining the static, no load, minimum bend radius) and store it at the structure until the splice enclosure has been installed. Place end caps on the exposed cable ends to prevent water penetration.